Incase you want to call each function without menu

Login();

Load();

Query();

Advanced_Search();

addAccount();

modifyAccount();

deleteAccount();

deleteMultiple();

changeStatus();

Print();

Withdraw();

Transfer();

Deposit();

Quit();

Save(); (auto implemented in other function)

dailyLimit(); (auto implemented in withdraw function)

Menu();

Report();

In menu to use a function from the selections type exactly as it says in the list

For example it says

-ADD

-LOGIN

-Advanced_Search

Just type advanced_search or login or add and it will run the chosen function

If you wont use menu function to call the other functions just call them in main.c…They are self-explanatory and will guide the user on what to do

To call a function you need to do it In main.c make sure the OurFunctions.h is defined and OurFunctions.c exists

```
"C:\Users\PC\OneDrive\Deskt
-LOGIN
-QUIT
login
Enter Username: 389sun
Enter Password: a
User not found
-LOGIN
-QUIT
login
Enter Username: aaa
Enter Password: 39sjm
Wrong Password
-LOGIN
-QUIT
quit
You have exited the program

Process returned 1 (0x1)   execution time : 18.779 s
Press any key to continue.
```

We will be testing all the functions in the menu function. This is when the user logs in multiples times and quits

```
-LOGIN
-QUIT
login
Enter Username: aaa
Enter Password: 123a
Access Granted
 Account Number: 9700000000 ||  Name: Michael Jones || Email: m.jones@gmail.com || Balance: 1000.00$ ||  Mobile: 01009700000 ||  Month: 12  || Year: 2007  ||  Status:  active

  Account Number: 9700000001 ||  Name: John Roberto || Email: j.roberto@outlook.com || Balance: 100.00$ ||  Mobile: 01009700001 ||  Month: 12  || Year: 2008  ||  Status:  active

  Account Number: 9700000002 ||  Name: Timothy Korman || Email: t.korman@gmail.com || Balance: 200.00$ ||  Mobile: 01009700002 ||  Month: 12  || Year: 2015  ||  Status:  active

  Account Number: 9700000003 ||  Name: Michael Robert || Email: michael@yahoo.com || Balance: 300.00$ ||  Mobile: 01009700003 ||  Month: 11  || Year: 2008  ||  Status:  inactive

  Account Number: 9700000004 ||  Name: Roberto Thomas || Email: rob.thomas@gmail.com || Balance: 400.50$ ||  Mobile: 01009700004 ||  Month: 11  || Year: 2015  ||  Status:  active

  Account Number: 9700000005 ||  Name: David Roberts || Email: david123@gmail.com || Balance: 400.50$ ||  Mobile: 01009700005 ||  Month: 10  || Year: 2015  ||  Status:  active

  Account Number: 9700000006 ||  Name: Daniel Graves || Email: dgrave@outlook.com || Balance: 450.00$ ||  Mobile: 01009700006 ||  Month: 1  || Year: 2020  ||  Status:  inactive

  Account Number: 9700000007 ||  Name: Philipe Brian || Email: p.brian@outlook.com || Balance: 460.00$ ||  Mobile: 01009700007 ||  Month: 2  || Year: 2020  ||  Status:  active

  Account Number: 9700000008 ||  Name: Adam Mark || Email: ad.mark@gmail.com || Balance: 350.00$ ||  Mobile: 0100970008 ||  Month: 10  || Year: 2015  ||  Status:  inactive

 -ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option:
```

If the user logs in Load function is automatically called and displays all the data

```
"C:\Users\PC\OneDrive\Deskt
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option: add
Enter account number: 30
Enter name: Abdullah
Enter email: Abdullah@gmail.com
Enter balance: 0
Enter mobile number: 01113635588
Save changes? (y/n): y
Account successfully added!
-ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option:
```

```
accounts.txt
File   Edit   View
9700000000,Michael Jones,m.jones@gmail.com,1000,01009700000,12-2007, active
9700000001,John Roberto,j.roberto@outlook.com,100,01009700001,12-2008, active
9700000002,Timothy Korman,t.korman@gmail.com,200,01009700002,12-2015, active
9700000003,Michael Robert,michael@yahoo.com,300,01009700003,11-2008, inactive
9700000004,Roberto Thomas,rob.thomas@gmail.com,400.5,01009700004,11-2015, active
9700000005,David Roberts,david123@gmail.com,400.5,01009700005,10-2015, active
9700000006,Daniel Graves,dgrave@outlook.com,450,01009700006,1-2020, inactive
9700000007,Philipe Brian,p.brian@outlook.com,460,01009700007,2-2020, active
9700000008,Adam Mark,ad.mark@gmail.com,350,0100970008,10-2015, inactive
9700000009,James Adams,j.adams@gmail.com,250,01009700009,5-2017, active
30,Abdullah,Abdullah@gmail.com,0.00,01113635588,12-2025, active
```

Test run for add func

```
"C:\Users\PC\OneDrive\Deskt
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option: modify
Enter account number: 30
Enter new name (current: Abdullah): Abdallah
Enter new email (current: Abdullah@gmail.com): a@gmail.com
Enter new mobile (current: 01113635588): 01113635588
Save changes? (y/n): y
Account 30 modified successfully
-ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option:
```

accounts.txt

```
9700000000,Michael Jones,m.jones@gmail.com,1000,01009700000,12-2007, active
9700000001,John Roberto,j.roberto@outlook.com,100,01009700001,12-2008, active
9700000002,Timothy Korman,t.korman@gmail.com,200,01009700002,12-2015, active
9700000003,Michael Robert,michael@yahoo.com,300,01009700003,11-2008, inactive
9700000004,Roberto Thomas,rob.thomas@gmail.com,400.5,01009700004,11-2015, active
9700000005,David Roberts,david123@gmail.com,400.5,01009700005,10-2015, active
9700000006,Daniel Graves,dgrave@outlook.com,450,01009700006,1-2020, inactive
9700000007,Philipe Brian,p.brian@outlook.com,460,01009700007,2-2020, active
9700000008,Adam Mark,ad.mark@gmail.com,350,0100970008,10-2015, inactive
9700000009,James Adams,j.adams@gmail.com,250,01009700009,5-2017, active
30,Abdallah,a@gmail.com,0.00,01113635588,12-2025, active
```

Test run for modify function

```
"C:\Users\PC\OneDrive\Deskt
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option: search
Account Number:9700000009
Name: James Adams
Email: j.adams@gmail.com
Balance: 250.00$
Mobile: 01009700009
Date Opened: May 2017
Status:  active

-ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option:
```

accounts.txt

```
9700000000,Michael Jones,m.jones@gmail.com,1000,01009700000,12-2007, active
9700000001,John Roberto,j.roberto@outlook.com,100,01009700001,12-2008, active
9700000002,Timothy Korman,t.korman@gmail.com,200,01009700002,12-2015, active
9700000003,Michael Robert,michael@yahoo.com,300,01009700003,11-2008, inactive
9700000004,Roberto Thomas,rob.thomas@gmail.com,400.5,01009700004,11-2015, active
9700000005,David Roberts,david123@gmail.com,400.5,01009700005,10-2015, active
9700000006,Daniel Graves,dgrave@outlook.com,450,01009700006,1-2020, inactive
9700000007,Philipe Brian,p.brian@outlook.com,460,01009700007,2-2020, active
9700000008,Adam Mark,ad.mark@gmail.com,350,0100970008,10-2015, inactive
9700000009,James Adams,j.adams@gmail.com,250,01009700009,5-2017, active
30,Abdallah,a@gmail.com,0.00,01113635588,12-2025, active
```

Test run for search function (Query)

```
"C:\Users\PC\OneDrive\Deskt
-REPORT
-PRINT
-QUIT
Choose an option: advanced_search
Enter Name: Roberto
Account Number: 9700000001
Name: John Roberto
Email: j.roberto@outlook.com
Balance: 100.00$
Mobile: 01009700001
Month: 12
Year: 2008
Status:  active

Account Number: 9700000004
Name: Roberto Thomas
Email: rob.thomas@gmail.com
Balance: 400.50$
Mobile: 01009700004
Month: 11
Year: 2015
Status:  active

-ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
```

accounts.txt

```
9700000000,Michael Jones,m.jones@gmail.com,1000,01009700000,12-2007, active
9700000001,John Roberto,j.roberto@outlook.com,100,01009700001,12-2008, active
9700000002,Timothy Korman,t.korman@gmail.com,200,01009700002,12-2015, active
9700000003,Michael Robert,michael@yahoo.com,300,01009700003,11-2008, inactive
9700000004,Roberto Thomas,rob.thomas@gmail.com,400.5,01009700004,11-2015, active
9700000005,David Roberts,david123@gmail.com,400.5,01009700005,10-2015, active
9700000006,Daniel Graves,dgrave@outlook.com,450,01009700006,1-2020, inactive
9700000007,Philipe Brian,p.brian@outlook.com,460,01009700007,2-2020, active
9700000008,Adam Mark,ad.mark@gmail.com,350,0100970008,10-2015, inactive
9700000009,James Adams,j.adams@gmail.com,250,01009700009,5-2017, active
30,Abdallah,a@gmail.com,0.00,01113635588,12-2025, active
```

Test run for Advanced_Search function

```
"C:\Users\PC\OneDrive\Deskt
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option: change_status
Enter account number: 9700000004
The current status: active

if you want to change the status enter '1' if not enter '2': 1
The status changed successfully
Save changes? (y/n): y
-ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option:
```

accounts.txt

```
9700000000,Michael Jones,m.jones@gmail.com,1000,01009700000,12-2007, active
9700000001,John Roberto,j.roberto@outlook.com,100,01009700001,12-2008, active
9700000002,Timothy Korman,t.korman@gmail.com,200,01009700002,12-2015, active
9700000003,Michael Robert,michael@yahoo.com,300,01009700003,11-2008, inactive
9700000004,Roberto Thomas,rob.thomas@gmail.com,400.50,01009700004,11-2015, inactive
9700000005,David Roberts,david123@gmail.com,400.5,01009700005,10-2015, active
9700000006,Daniel Graves,dgrave@outlook.com,450,01009700006,1-2020, inactive
9700000007,Philipe Brian,p.brian@outlook.com,460,01009700007,2-2020, active
9700000008,Adam Mark,ad.mark@gmail.com,350,0100970008,10-2015, inactive
9700000009,James Adams,j.adams@gmail.com,250,01009700009,5-2017, active
30,Abdallah,a@gmail.com,0.00,01113635588,12-2025, active
```

Test run for change_status function

```
"C:\Users\PC\OneDrive\Deskt
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option: deposit
Please enter account number: 30
Please enter the amount you wish to deposit (Balance: 0.00)(maximum $10,000): 1000
Transaction Successful! Your balance is  1000.00$
Save changes? (y/n): y
-ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option:
```

accounts.txt

```
9700000000,Michael Jones,m.jones@gmail.com,1000,01009700000,12-2007, active
9700000001,John Roberto,j.roberto@outlook.com,100,01009700001,12-2008, active
9700000002,Timothy Korman,t.korman@gmail.com,200,01009700002,12-2015, active
9700000003,Michael Robert,michael@yahoo.com,300,01009700003,11-2008, inactive
9700000004,Roberto Thomas,rob.thomas@gmail.com,400.50,01009700004,11-2015, inactive
9700000005,David Roberts,david123@gmail.com,400.5,01009700005,10-2015, active
9700000006,Daniel Graves,dgrave@outlook.com,450,01009700006,1-2020, inactive
9700000007,Philipe Brian,p.brian@outlook.com,460,01009700007,2-2020, active
9700000008,Adam Mark,ad.mark@gmail.com,350,0100970008,10-2015, inactive
9700000009,James Adams,j.adams@gmail.com,250,01009700009,5-2017, active
30,Abdallah,a@gmail.com,1000.00,01113635588,12-2025, active
```

30.txt

```
Deposit,1000,27-12-2025
```

Test run for deposit function

Deposit,amount deposited ,date

```
"C:\Users\PC\OneDrive\Deskt
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option: withdraw
Please enter account number: 30
Please enter the amount you wish to withdraw (Balance: 1000.00)(maximum $10,000): 500
Transaction Successful! Your new balance is  500.00$
Save changes? (y/n): y
-ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option:
```

accounts.txt

```
9700000000,Michael Jones,m.jones@gmail.com,1000,01009700000,12-2007, active
9700000001,John Roberto,j.roberto@outlook.com,100,01009700001,12-2008, active
9700000002,Timothy Korman,t.korman@gmail.com,200,01009700002,12-2015, active
9700000003,Michael Robert,michael@yahoo.com,300,01009700003,11-2008, inactive
9700000004,Roberto Thomas,rob.thomas@gmail.com,400.50,01009700004,11-2015, inactive
9700000005,David Roberts,david123@gmail.com,400.5,01009700005,10-2015, active
9700000006,Daniel Graves,dgrave@outlook.com,450,01009700006,1-2020, inactive
9700000007,Philipe Brian,p.brian@outlook.com,460,01009700007,2-2020, active
9700000008,Adam Mark,ad.mark@gmail.com,350,0100970008,10-2015, inactive
9700000009,James Adams,j.adams@gmail.com,250,01009700009,5-2017, active
30,Abdallah,a@gmail.com,500.00,01113635588,12-2025, active
```

30.txt

```
Deposit,1000,27-12-2025
Withdraw,500,27-12-2025
```

Test run for withdraw function

Withdraw,amount withdrawn,date

```
Choose an option: print
How do you want the print to be sorted? (name,balance,date): name
Account Number: 30 || Name: Abdallah || Email: a@gmail.com || Balance: 500.00$ || Mobile: 01113635588 || Month: 12  || Year: 2025  || Status:  active

Account Number: 9700000008 || Name: Adam Mark || Email: ad.mark@gmail.com || Balance: 350.00$ || Mobile: 0100970008 || Month: 10  || Year: 2015  || Status:  inactive

Account Number: 9700000006 || Name: Daniel Graves || Email: dgrave@outlook.com || Balance: 450.00$ || Mobile: 01009700006 || Month: 1  || Year: 2020  || Status:  inactive

Account Number: 9700000005 || Name: David Roberts || Email: david123@gmail.com || Balance: 400.50$ || Mobile: 01009700005 || Month: 10  || Year: 2015  || Status:  active

Account Number: 9700000009 || Name: James Adams || Email: j.adams@gmail.com || Balance: 250.00$ || Mobile: 01009700009 || Month: 5  || Year: 2017  || Status:  active

Account Number: 9700000001 || Name: John Roberto || Email: j.roberto@outlook.com || Balance: 100.00$ || Mobile: 01009700001 || Month: 12  || Year: 2008  || Status:  active

Account Number: 9700000000 || Name: Michael Jones || Email: m.jones@gmail.com || Balance: 1000.00$ || Mobile: 01009700000 || Month: 12  || Year: 2007  || Status:  active

Account Number: 9700000003 || Name: Michael Robert || Email: michael@yahoo.com || Balance: 300.00$ || Mobile: 01009700003 || Month: 11  || Year: 2008  || Status:  inactive

Account Number: 9700000007 || Name: Philipe Brian || Email: p.brian@outlook.com || Balance: 460.00$ || Mobile: 01009700007 || Month: 2  || Year: 2020  || Status:  active

Account Number: 9700000004 || Name: Roberto Thomas || Email: rob.thomas@gmail.com || Balance: 400.50$ || Mobile: 01009700004 || Month: 11  || Year: 2015  || Status:  inactive

Account Number: 9700000002 || Name: Timothy Korman || Email: t.korman@gmail.com || Balance: 200.00$ || Mobile: 01009700002 || Month: 12  || Year: 2015  || Status:  active
```

Test run for print sort (name)

```
Choose an option: print
How do you want the print to be sorted? (name,balance,date): balance
Account Number: 9700000001 || Name: John Roberto || Email: j.roberto@outlook.com || Balance: 100.00$ || Mobile: 01009700001 || Month: 12  || Year: 2008  || Status:  active

Account Number: 9700000002 || Name: Timothy Korman || Email: t.korman@gmail.com || Balance: 200.00$ || Mobile: 01009700002 || Month: 12  || Year: 2015  || Status:  active

Account Number: 9700000009 || Name: James Adams || Email: j.adams@gmail.com || Balance: 250.00$ || Mobile: 01009700009 || Month: 5  || Year: 2017  || Status:  active

Account Number: 9700000003 || Name: Michael Robert || Email: michael@yahoo.com || Balance: 300.00$ || Mobile: 01009700003 || Month: 11  || Year: 2008  || Status:  inactive

Account Number: 9700000008 || Name: Adam Mark || Email: ad.mark@gmail.com || Balance: 350.00$ || Mobile: 0100970008 || Month: 10  || Year: 2015  || Status:  inactive

Account Number: 9700000004 || Name: Roberto Thomas || Email: rob.thomas@gmail.com || Balance: 400.50$ || Mobile: 01009700004 || Month: 11  || Year: 2015  || Status:  inactive

Account Number: 9700000005 || Name: David Roberts || Email: david123@gmail.com || Balance: 400.50$ || Mobile: 01009700005 || Month: 10  || Year: 2015  || Status:  active

Account Number: 9700000006 || Name: Daniel Graves || Email: dgrave@outlook.com || Balance: 450.00$ || Mobile: 01009700006 || Month: 1  || Year: 2020  || Status:  inactive

Account Number: 9700000007 || Name: Philipe Brian || Email: p.brian@outlook.com || Balance: 460.00$ || Mobile: 01009700007 || Month: 2  || Year: 2020  || Status:  active

Account Number: 30 || Name: Abdallah || Email: a@gmail.com || Balance: 500.00$ || Mobile: 01113635588 || Month: 12  || Year: 2025  || Status:  active

Account Number: 9700000000 || Name: Michael Jones || Email: m.jones@gmail.com || Balance: 1000.00$ || Mobile: 01009700000 || Month: 12  || Year: 2007  || Status:  active
```

Test run for print sort (balance)

```
Choose an option: print
How do you want the print to be sorted? (name,balance,date): date
Account Number: 9700000000 || Name: Michael Jones || Email: m.jones@gmail.com || Balance: 1000.00$ || Mobile: 01009700000 || Month: 12  || Year: 2007  || Status:  active

Account Number: 9700000003 || Name: Michael Robert || Email: michael@yahoo.com || Balance: 300.00$ || Mobile: 01009700003 || Month: 11  || Year: 2008  || Status:  inactive

Account Number: 9700000001 || Name: John Roberto || Email: j.roberto@outlook.com || Balance: 100.00$ || Mobile: 01009700001 || Month: 12  || Year: 2008  || Status:  active

Account Number: 9700000005 || Name: David Roberts || Email: david123@gmail.com || Balance: 400.50$ || Mobile: 01009700005 || Month: 10  || Year: 2015  || Status:  active

Account Number: 9700000008 || Name: Adam Mark || Email: ad.mark@gmail.com || Balance: 350.00$ || Mobile: 0100970008 || Month: 10  || Year: 2015  || Status:  inactive

Account Number: 9700000004 || Name: Roberto Thomas || Email: rob.thomas@gmail.com || Balance: 400.50$ || Mobile: 01009700004 || Month: 11  || Year: 2015  || Status:  inactive

Account Number: 9700000002 || Name: Timothy Korman || Email: t.korman@gmail.com || Balance: 200.00$ || Mobile: 01009700002 || Month: 12  || Year: 2015  || Status:  active

Account Number: 9700000009 || Name: James Adams || Email: j.adams@gmail.com || Balance: 250.00$ || Mobile: 01009700009 || Month: 5  || Year: 2017  || Status:  active

Account Number: 9700000006 || Name: Daniel Graves || Email: dgrave@outlook.com || Balance: 450.00$ || Mobile: 01009700006 || Month: 1  || Year: 2020  || Status:  inactive

Account Number: 9700000007 || Name: Philipe Brian || Email: p.brian@outlook.com || Balance: 460.00$ || Mobile: 01009700007 || Month: 2  || Year: 2020  || Status:  active

Account Number: 30 || Name: Abdallah || Email: a@gmail.com || Balance: 500.00$ || Mobile: 01113635588 || Month: 12  || Year: 2025  || Status:  active
```

Test run for print sort (date)

```
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option: transfer
Please Enter Sender Account Number: 30
Please Enter Reciever Account Number: 9700000007
Enter the amount you wish to send (Sender's Balance: 500.00$)(Reciever's Balance: 460.00$): 40
Save changes? (y/n): y
Transaction Successful!
-ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option:
```

accounts.txt

```
9700000000,Michael Jones,m.jones@gmail.com,1000,01009700000,12-2007, active
9700000001,John Roberto,j.roberto@outlook.com,100,01009700001,12-2008, active
9700000002,Timothy Korman,t.korman@gmail.com,200,01009700002,12-2015, active
9700000003,Michael Robert,michael@yahoo.com,300,01009700003,11-2008, inactive
9700000004,Roberto Thomas,rob.thomas@gmail.com,400.50,01009700004,11-2015, inactive
9700000005,David Roberts,david123@gmail.com,400.5,01009700005,10-2015, active
9700000006,Daniel Graves,dgrave@outlook.com,450,01009700006,1-2020, inactive
9700000007,Philipe Brian,p.brian@outlook.com,500.000000,01009700007,2-2020, active
9700000008,Adam Mark,ad.mark@gmail.com,350,0100970008,10-2015, inactive
9700000009,James Adams,j.adams@gmail.com,250,01009700009,5-2017, active
30,Abdallah,a@gmail.com,460.000000,01113635588,12-2025, active
```

30.txt

```
Deposit,1000,27-12-2025
Withdraw,500,27-12-2025
Transfer,30,9700000007,-40.000000,27-12-2025
```

9700000007.txt

```
Transfer,30,9700000007,+40.000000,27-12-2025
```

Test run for transaction function
Transfer,sender,receiver,balance sent,date

+ means received, - means sent

```
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option: report
Enter account number: 30
Deposit,1000,27-12-2025
Withdraw,500,27-12-2025
Transfer,30,9700000007,-40.000000,27-12-2025
-ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option:
```

30.txt

```
Deposit,1000,27-12-2025
Withdraw,500,27-12-2025
Transfer,30,9700000007,-40.000000,27-12-2025
```

Test run for Report function

```
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option: delete
Enter account number: 30
Save changes? (y/n): y
Account 30 deleted successfully!
-ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option:
```

accounts.txt

```
9700000000,Michael Jones,m.jones@gmail.com,1000,01009700000,12-2007, active
9700000001,John Roberto,j.roberto@outlook.com,100,01009700001,12-2008, active
9700000002,Timothy Korman,t.korman@gmail.com,200,01009700002,12-2015, active
9700000003,Michael Robert,michael@yahoo.com,300,01009700003,11-2008, inactive
9700000004,Roberto Thomas,rob.thomas@gmail.com,400.50,01009700004,11-2015, inactive
9700000005,David Roberts,david123@gmail.com,400.5,01009700005,10-2015, active
9700000006,Daniel Graves,dgrave@outlook.com,450,01009700006,1-2020, inactive
9700000007,Philipe Brian,p.brian@outlook.com,500.000000,01009700007,2-2020, active
9700000008,Adam Mark,ad.mark@gmail.com,350,0100970008,10-2015, inactive
9700000009,James Adams,j.adams@gmail.com,250,01009700009,5-2017, active
```

Test run for delete function (I set the balance of that account to zero to be able to demonstrate)

accounts.txt

```
9700000000,Michael Jones,m.jones@gmail.com,1000,01009700000,12-2007, active
9700000001,John Roberto,j.roberto@outlook.com,100,01009700001,12-2008, active
9700000002,Timothy Korman,t.korman@gmail.com,0,01009700002,12-2015, active
9700000003,Michael Robert,michael@yahoo.com,300,01009700003,11-2008, inactive
9700000004,Roberto Thomas,rob.thomas@gmail.com,0,01009700004,10-2015, inactive
9700000005,David Roberts,david123@gmail.com,400.5,01009700005,10-2015, active
9700000006,Daniel Graves,dgrave@outlook.com,450,01009700006,10-2015, inactive
9700000007,Philipe Brian,p.brian@outlook.com,500.000000,01009700007,2-2020, active
9700000008,Adam Mark,ad.mark@gmail.com,0,0100970008,10-2015, inactive
9700000009,James Adams,j.adams@gmail.com,250,01009700009,5-2017, active
```

```
"C:\Users\PC\OneDrive\Deskt
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option: delete_multiple
Delete multiple accounts by (activity,date): date
Enter date of accounts you want deleted (YYYY-MM): 2015-10
Save changes? (y/n): y
Account 9700000004 deleted successfully!
Account 9700000008 deleted successfully!
-ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option:
```

accounts.txt

```
9700000000,Michael Jones,m.jones@gmail.com,1000,01009700000,12-2007, active
9700000001,John Roberto,j.roberto@outlook.com,100,01009700001,12-2008, active
9700000002,Timothy Korman,t.korman@gmail.com,0,01009700002,12-2015, active
9700000003,Michael Robert,michael@yahoo.com,300,01009700003,11-2008, inactive
9700000005,David Roberts,david123@gmail.com,400.5,01009700005,10-2015, active
9700000006,Daniel Graves,dgrave@outlook.com,450,01009700006,10-2015, inactive
9700000007,Philipe Brian,p.brian@outlook.com,500.000000,01009700007,2-2020, active
9700000009,James Adams,j.adams@gmail.com,250,01009700009,5-2017, active
```

Test run for delete_multiple by date function

The above txt is before deletion the other one is after

accounts.txt

```
9700000000,Michael Jones,m.jones@gmail.com,1000,01009700000,12-2007, active
9700000001,John Roberto,j.roberto@outlook.com,100,01009700001,12-2008, active
9700000002,Timothy Korman,t.korman@gmail.com,0,01009700002,12-2015, active
9700000003,Michael Robert,michael@yahoo.com,300,01009700003,11-2008, inactive
9700000005,David Roberts,david123@gmail.com,0,01009700005,12-2025, inactive
9700000006,Daniel Graves,dgrave@outlook.com,0,01009700006,10-2015, inactive
9700000007,Philipe Brian,p.brian@outlook.com,0.000000,01009700007,2-2020, inactive
9700000009,James Adams,j.adams@gmail.com,0,01009700009,5-2017, inactive
```

```
"C:\Users\PC\OneDrive\Deskt
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option: delete_multiple
Delete multiple accounts by (activity,date): activity
Save changes? (y/n): y
Account 9700000006 deleted successfully!
Account 9700000007 deleted successfully!
Account 9700000009 deleted successfully!
-ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option:
```

accounts.txt

```
9700000000,Michael Jones,m.jones@gmail.com,1000,01009700000,12-2007, active
9700000001,John Roberto,j.roberto@outlook.com,100,01009700001,12-2008, active
9700000002,Timothy Korman,t.korman@gmail.com,0,01009700002,12-2015, active
9700000003,Michael Robert,michael@yahoo.com,300,01009700003,11-2008, inactive
9700000005,David Roberts,david123@gmail.com,0,01009700005,12-2025, inactive
```

Test run for delete_multiple by activity function

The above txt is before deletion the other one is after

```
"C:\Users\PC\OneDrive\Deskt
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option: delete_multiple
Delete multiple accounts by (activity,date): activity
Save changes? (y/n): y
Account 9700000006 deleted successfully!
Account 9700000007 deleted successfully!
Account 9700000009 deleted successfully!
-ADD
-DELETE
-DELETE_MULTIPLE
-MODIFY
-SEARCH
-ADVANCED_SEARCH
-CHANGE_STATUS
-WITHDRAW
-DEPOSIT
-TRANSFER
-REPORT
-PRINT
-QUIT
Choose an option: quit
You have exited the program

Process returned 1 (0x1)   execution time : 2865.653 s
Press any key to continue.
```

Test function for quit

Notes:

Only algorithms used were bubble sort in the helper functions sortByDate and sortByBalance and sortByName

In any function that overwrites data the method was to make a temp txt file and then write all the updated data there..after that we rename it accounts.txt and delete the original accounts.txt file (ofcourse after saving changes if discarded it wont)

Time.h was used to gain realtime data about the date for dailyLimit and add and transaction functions

To add colours we used codes such as \033[31m which is for red if u change the 1 to (1-10) it will change colors \033[30m is put at the end of printf to reset the colors (if this isn't included the texts will all be the color of the previous code) think of it as a closing tag like <html></html>